Curriculum Vitae

Education

Medical Engineering MSc – Ø 1.9

From 10/2016 to 08/2019 at FAU Erlangen-Nuremberg

Study track: Medical Image & Data Processing

Master's Thesis: *'Development of a flexible 3D-Visualisation for a Simulation Software in Medical Robotics'*

- Software Development and Design Patterns
- Visualisation in Unity, VTK and Augmented Reality
- Processing of RGB-D-data for room capture
- Evaluation with a user study

Semester Abroad

From 08/2018 to 12/2018 at University of La Serena, Chile

- Web Development and Usability Engineering
- Project Evaluation

Medical Engineering B.Eng. – Ø 1.7

From 10/2012 to 07/2016 at Nuremberg Institute of Technology

Study track: Electrical Engineering & Information Technology

Bachelor's Thesis: *'Optimisation of an Algorithm for Motion Compensation in Digital Subtraction Angiography through Coordinate Transformation'*

- 2D-2D-Registration of X-Ray images
- Evaluation on clinical data

Matriculation Exam – Ø 1.8

06/2011 at Eckental Grammar School

Student Projects

Detection and Segmentation of Medieval Notary Certificates with Deep Learning

Research Internship from 04/2019 to 08/2019 at Pattern Recognition Lab, FAU Erlangen-Nuremberg

A Future-Orientated Network Infrastructure for Pedagogic Networks at Schools

Project Work from 02/2015 to 07/2015 at City of Nuremberg

Martin Leipert

Medical Engineering MSc

Personal Data

01.10.1993

in Forchheim

German

Contact Information

+49 1577 8309855

martin.leipert@t-online.de

Hopfenstraße 22
D-91322 Gräfenberg

martinleipert.github.io

Focus

- Medical Image Processing
- Computer Vision
- Machine Learning
- Numerical Mathematics
- Computer Graphics
- Software Development

Curriculum Vitae

Martin Leipert

Professional Career

Working Student
Since 05/2019 at Siemens Healthineers, Forchheim

- Augmented Reality with the Microsoft HoloLens and Unity
- Development of a kinematic robotic simulation in C#

Working Student
From 05/2016 to 11/2017 at Siemens Healthineers, Forchheim

- Plug-in development for a kinematic robotic simulation in C#
- Computer vision with x-ray systems
- Development of a dynamics simulation in C++
- Automation for accuracy measurements in robot calibration

Intern & Working Student
From 09/2014 to 09/2015 at Siemens Sector Healthcare, Forchheim

- Tools for evaluation of robot calibration accuracy
- Automated analysis of angiography system logfiles

Tutor for Electrical Engineering Fundamentals
From 10/2014 to 01/2015 at Nuremberg Institute of Technology

Working Student
From 04/2014 to 08/2014 at Siemens Sector Healthcare, Forchheim

- Evaluation of an algorithm for motion compensation in Digital Subtraction Angiography on clinical data

Publications

A Mixed Reality Simulation for Robotic Systems
Leipert M., Jenny Sadowski, Kießling M., Kwemou-Ngandeu M., Maier A.
Bildverarbeitung für die Medizin 2019, Wiesbaden, Springer Fachmedien, 2019

Gräfenberg, 21. August 2019

Martin Leipert

Awards

Faculty Award of the Faculty efi 2016
Best Medical Engineering graduate at TH Nuremberg in the year 2016

Most Innovative Project
Innovation Research Lab Exhibition 2016 of ZIMT at FAU Erlangen

Programming Skills

C#
From: Master's Thesis, Working Student

C++
From: Working Student, Studies

Python
From: Working Student

Matlab
From: Working Student, Studies

Java
From: Studies

Web Development
From: Semester Abroad, Homepage Supervision

Additional: Unity, PyTorch

Language Skills

German
Native

English
Fluent

Spanish
Advanced

Voluntary Activity

Social Democratic Party of Germany
Member of the local executive board